9 **(a)** State what is meant by a *field of force*.

A field of force (or force field) is a region of space in which a particle experiences a force due to a physical property that it possesses. [1]

(b) Two parallel metal plates are separated by a distance of 6.0 cm in a vacuum, as shown in Fig. 9.1. The plates have length 16 cm and potential difference of 2400 V.

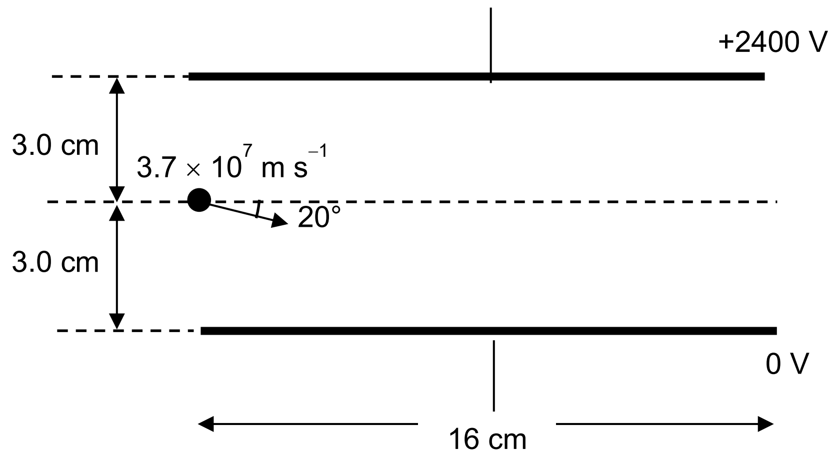


Fig. 9.1

An electron with speed 3.7×10^7 m s^{-1} enters the region between the plates. The initial direction of the electron is 20° below the midline between the plates.

(i) Calculate the acceleration of the electron and state its direction.

$$F = qE = q\frac{\Delta V}{d}$$

$$= \left(1.60 \times 10^{-19}\right)\frac{2400}{0.06}$$

$$= 6.40 \times 10^{-15} \text{ [1]}$$

$$a = \frac{F}{m}$$

$$= \frac{6.40 \times 10^{-15}}{9.11 \times 10^{-31}}$$

$$= 7.0252 \times 10^{15} \text{ upwards [2]}$$

acceleration = ... m s^{-2}

direction = ...

[3]

(ii) Calculate the time taken for the electron to reach the other end of the plate.

$$t = \frac{s_x}{u_x}$$
$$= \frac{0.16}{3.7 \times 10^7 \cos 20°}$$
$$= 4.6018 \times 10^{-9}\,\text{s}$$
$$= 4.60 \times 10^{-9}\,\text{s}$$

time = ... s [1]

(iii) Use your answers in **(b)(i)** and **(ii)** to determine whether the electron will collide with any metal plate as it passes through the region between the plates.

Considering whether electron collides with top plate:

$$s_y = u_y t + \frac{1}{2} a_y t^2$$
$$= \left(3.7 \times 10^7 \sin 20°\right)\left(4.6018 \times 10^{-9}\right) + \frac{1}{2}\left(-7.0252 \times 10^{15}\right)\left(4.6018 \times 10^{-9}\right)^2$$
$$= -0.0162\ \text{m}\ \ [1]\ \text{which is less than}\ -0.030\ \text{m}$$

Considering whether electron collides with bottom plate:
when $v_y = 0$:

$$0 = u_y + a_y t$$
$$t = \frac{-\left(3.7 \times 10^7 \sin 20°\right)}{-7.0252 \times 10^{15}}$$
$$= 1.8013 \times 10^{-9}\ [1]$$
$$s_y = u_y t + \frac{1}{2} a_y t^2$$
$$= \left(3.7 \times 10^7 \sin 20°\right)\left(1.8013 \times 10^{-9}\right) + \frac{1}{2}\left(-7.0252 \times 10^{15}\right)\left(1.8013 \times 10^{-9}\right)^2$$
$$= 0.0114\ \text{m}\ [1]\ \text{which is less than}\ 0.030\ \text{m}$$

Therefore, the electron will not collide with neither the top nor bottom plate. [1]

[3]

(iv) Hence, sketch, on Fig. 9.1, the path of the electron. [1]

(iv) Describe the path of the electron in the field.

For an electron in a uniform electric field, it is a parabolic path.
....[1]

(c) Another electron of the same speed now enters a region of uniform magnetic field of flux density 4.5 mT as shown in Fig. 9.2.

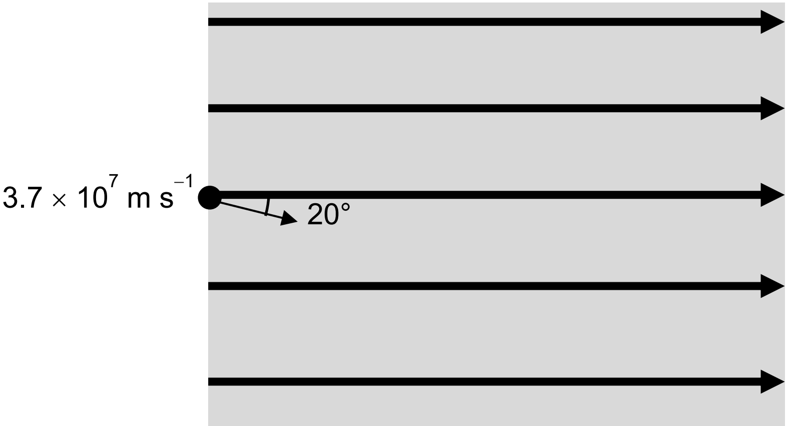


Fig. 9.2

The initial direction of the electron is at an angle of 20° to the direction of magnetic field.

(i) When the electron enters the magnetic field, the component of its velocity $v_\perp$ normal to the direction of the magnetic field causes the electron to begin to follow a circular path. Explain why.

When the velocity of the electron is perpendicular to the magnetic field, it gives rise to a magnetic force which is always perpendicular to its velocity [1]. This magnetic force on the moving charge provides a centripetal force [1] for it to move in a uniform circular motion. [2]

(ii) Calculate the radius of this circular path.

By Newton's Second Law, $\sum F = ma_c$

$$Bqv_\perp = \frac{mv_\perp^2}{r} \quad [1]$$

$$(4.5\times10^{-3})(1.60\times10^{-19}) = \frac{(9.11\times10^{-31})(3.7\times10^{7}\sin 20°)}{r}$$

$$r = 0.0160 \text{ m} \quad [1]$$

Correct component of velocity [1]

radius = ... m [3]

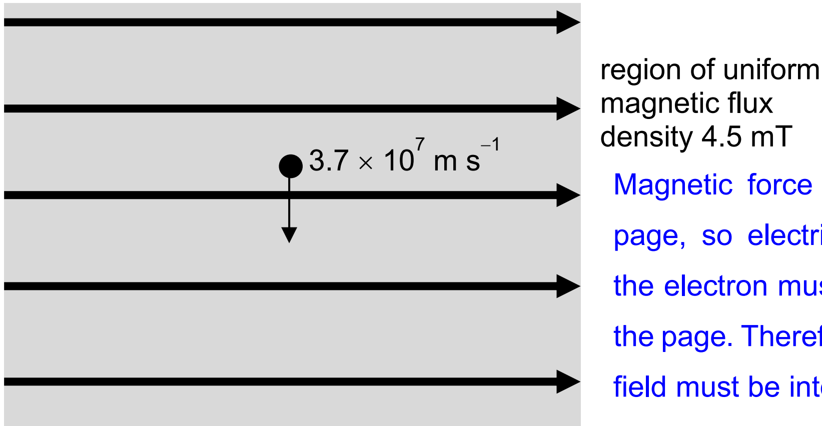
(iii) State the magnitude of the force on the electron due to the component of its velocity along the direction of the field.

Zero. [1]

(iv) Use your answers in **(c)(ii)** and **(iii)** to describe the resultant path of the electron in the field.

For an electron moving at an angle to a uniform magnetic field, it is a helical path. [1]

(d) Another electron of the same speed is projected downwards in the magnetic field as shown in Fig. 9.3. A uniform electric field is now switched on in the same region as the magnetic field. The magnitude of the electric field is adjusted so that the electron moves undeviated through the two fields.



Magnetic force is into the page, so electric force on the electron must be out of the page. Therefore electric field must be into the page.

Draw crosses [1]

Fig. 9.3

(i) On Fig. 9.3, draw the direction of the electric field. [1]

(ii) Determine the magnitude E of the electric field strength.

By Newton's First Law, $\sum F = 0$

$$Bqv = qE \quad [1]$$

$$\left(4.5\times10^{-3}\right)\left(3.7\times10^{7}\right) = E$$

$$E = 1.67\times10^{5} \text{ V/m} \quad [1]$$

E = ... V m^{-1} [2]

[Total: 20]